INCH-POUND

MIL-C-19978/12A
16 February 1995
SUPERSEDING
MIL-C-19978/12
20 October 1972

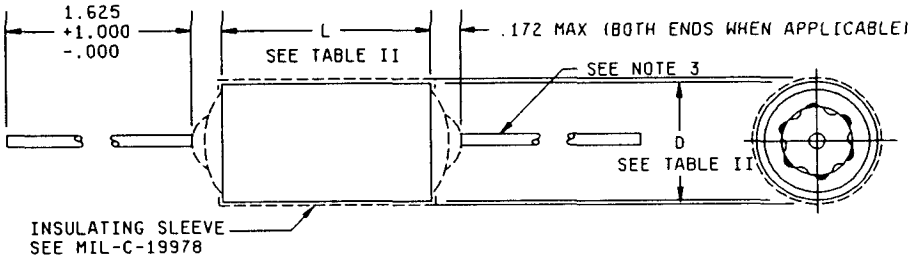
MILITARY SPECIFICATION

CAPACITORS, FIXED, PLASTIC (OR PAPER-PLASTIC) DIELECTRIC, AXIAL-WIRE TERMINAL, TUBULAR (INSULATED) (HERMETICALLY SEALED IN METAL CASES), ESTABLISHED RELIABILITY, STYLE CQR19

INACTIVE FOR NEW EQUIPMENT DESIGN AFTER 20 OCTOBER 1972.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-19978



Inches	mm	Inches	mm
.172	4.37	1.625	41.28
1.000	25.40	2.625	66.67

NOTES:
1. Dimensions are inches.
2. Metric equivalents are given for general information only.
3. Leads shall be of tinned solid wire .025 (0.64 mm) (No. 22 AWG) for cases .235 (5.97 mm) through .312 (7.92 mm) diameter; and .032 (0.81 mm) (No. 20 AWG) for cases .400 (10.16 mm) diameter and above. Tolerance on all lead wire diameters shall be +.004, -.001 (+0.10, -0.03 mm).
4. Capacitors with dimension L of 1.562 (39.67 mm) or D of .562 (14.27 mm) and larger are not intended to be supported by their leads. These capacitors shall be supported with a supplementary means of mounting, such as a wrap-around band. The supporting device will not be supplied with the capacitor.
5. Lead length may be a minimum of 1.00 inch long for use in tape and reel packaging when specified in the ordering data.

FIGURE 1. Style CQR19 capacitors.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1 and table I.

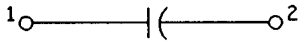
Case material: Nonmagnetic (end seal may be of magnetic material).

Dielectric material: Paper and polyethylene terephthalate (K).

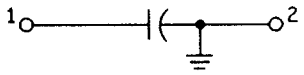
Terminals: Axial-wire lead (see figure 1).

Circuit diagram:

CIRCUIT DIAGRAM - 1



3



Rated voltage: See table I.

Rated temperature: -65°C to +125°C.

Capacitance (Cap.) (nom): See table I.

Capacitance tolerance: See table I.

Dissipation factor (DF) (maximum): 1.0 percent.

Failure rate level: In accordance with MIL-C-19978.

Reliability: In accordance with MIL-C-19978.

Burn-in: In accordance with MIL-C-19978.

Radiographic inspection: In accordance with MIL-C-19978.

Seal: Method 112 of MIL-STD-202, test condition letter A.

Dielectric withstanding voltage (DWV):

Sleeving: In accordance with MIL-C-19978.

Barometric pressure: In accordance with MIL-C-19978.

Test points:

Circuit diagram 1: Between terminals and case.

Circuit diagram 3: Between ungrounded terminal and case.

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TABLE I. Style CQR19 capacitors.

Part or Identifying Number (PIN) 1/	DC voltage rating	Capacitance rating	Capacitance tolerance	Failure rate level	Case dimensions 2/			
					Circuit 1		Circuit 3	
					L ±.031 (0.79)	D +.015 (0.38) -.005 (0.13)	L ±.031 (0.79)	D +.015 (0.38) -.005 (0.13)
	Volts	µF			Inches mm	Inches mm	Inches mm	Inches mm
CQR19A-KC124-3-	200	.12	J, K	M,P,R,S	1.125 (28.58)	.562 (14.27)	1.063 (27.00)	.562 (14.28)
CQR19A-KC154-3-	200	.15	J, K	|	1.125 (28.58)	.562 (14.27)	1.063 (27.00)	.562 (14.28)
CQR19A-KD392-3-	300	.0039	J, K	|	.813 (20.65)	.235 (5.97)	.719 (18.26)	.235 (5.97)
CQR19A-KD472-3-	300	.0047	J, K	|	.813 (20.65)	.235 (5.97)	.719 (18.26)	.235 (5.97)
CQR19A-KD183-3-	300	.018	J, K	|	1.062 (26.97)	.312 (7.92)	.969 (24.61)	.312 (7.92)
CQR19A-KD223-3-	300	.022	J, K	|	1.062 (26.97)	.312 (7.92)	.969 (24.61)	.312 (7.92)
CQR19A-KD563-3-	300	.056	J, K	|	1.312 (33.32)	.400 (10.16)	1.219 (30.96)	.400 (10.16)
CQR19A-KD683-3-	300	.068	J, K	|	1.312 (33.32)	.400 (10.16)	1.219 (30.96)	.400 (10.16)
CQR19A-KD394-3-	300	.39	J, K	|	1.875 (47.63)	.750 (19.05)	1.812 (46.02)	.750 (19.05)
CQR19A-KD474-3-	300	.47	J, K	|	1.875 (47.63)	.750 (19.05)	1.812 (46.02)	.750 (19.05)
CQR19A-KD564-3-	300	.56	J, K	|	2.375 (60.33)	.750 (19.05)	2.312 (58.72)	.750 (19.05)
CQR19A-KD684-3-	300	.68	J, K	|	2.375 (60.33)	.750 (19.05)	2.312 (58.72)	.750 (19.05)
CQR19A-KD824-3-	300	.82	J, K	|	2.125 (53.98)	1.000 (25.40)	2.062 (52.37)	1.000 (25.40)
CQR19A-KD105-3-	300	1.0	J, K	|	2.125 (53.98)	1.000 (25.40)	2.062 (52.37)	1.000 (25.40)
CQR19A-KE184-3-	400	.18	J, K	|	1.500 (38.10)	.670 (17.02)	1.438 (36.53)	.670 (17.02)
CQR19A-KE224-3-	400	.22	J, K	|	1.500 (38.10)	.670 (17.02)	1.438 (36.53)	.670 (17.02)
CQR19A-KE564-3-	400	.56	J, K	|	2.625 (66.68)	.750 (19.05)	2.562 (65.07)	.750 (19.05)
CQR19A-KE684-3-	400	.68	J, K	|	2.625 (66.68)	.750 (19.05)	2.562 (65.07)	.750 (19.05)
CQR19A-KE824-3-	400	.82	J, K	|	2.250 (57.15)	1.000 (25.40)	2.188 (55.58)	1.000 (25.40)
CQR19A-KE105-3-	400	1.0	J, K	|	2.250 (57.15)	1.000 (25.40)	2.188 (55.58)	1.000 (25.40)
CQR19A-KF273-3-	600	.027	J, K	|	1.562 (39.67)	.400 (10.16)	1.469 (37.31)	.400 (10.16)
CQR19A-KF333-3-	600	.033	J, K	|	1.562 (39.67)	.400 (10.16)	1.469 (37.31)	.400 (10.16)
CQR19A-KF393-3-	600	.039	J, K	|	1.125 (28.58)	.562 (14.27)	1.062 (26.97)	.562 (14.28)
CQR19A-KF473-3-	600	.047	J, K	|	1.125 (28.58)	.562 (14.27)	1.062 (26.97)	.562 (14.28)
CQR19A-KF394-3-	600	.39	J, K	|	2.125 (53.98)	1.000 (25.40)	2.062 (52.37)	1.000 (25.40)
CQR19A-KF474-3-	600	.47	J, K	M,P,R,S	2.125 (53.98)	1.000 (25.40)	2.062 (52.37)	1.000 (25.40)

1/ Complete PIN shall include symbols to indicate circuit, capacitance tolerance, and failure rate level, as applicable.
2/ Dimensions are for basic case; sleeving shall extend .016 inch (0.41 mm) minimum, .062 inch (1.57 mm) maximum beyond each end of the capacitor body; however, if a shrink-fitted insulation is used for the sleeving, it shall lap over the ends of the capacitor body. Add .047 inch (1.19 mm) maximum to the nominal for capacitor diameter.

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Insulation resistance (IR):

Sleeving: In accordance with MIL-C-19978.

Terminal to terminal: See table II.

TABLE II. Terminal-to-terminal insulation resistance.

Capacitance rating	Minimum insulation resistance
Characteristic K	
	At 25°C
0 to 0.6 microfarad	25,000 megohms
Greater than 0.6 microfarad	15,000 megohm-microfarads 1/
	At 125°C
0 to 0.08 microfarad	250 megohms
Greater than 0.08 microfarad	20 megohm-microfarads 1/

1/ Product obtained by multiplying the capacitance in microfarads by the insulation resistance in megohms.

Terminal to case: Greater than 10,000 megohms.

Vibration, high frequency: Method 204 of MIL-STD-202, test condition B, with the following exception:

Direction and duration of motion: 4 hours in each of two mutually perpendicular directions (total of 8 hours), one parallel and the other perpendicular to the cylindrical axis.

Salt spray: In accordance with MIL-C-19978.

Immersion:

DWV:

Sleeving: In accordance with MIL-C-19978.

IR:

Sleeving: In accordance with MIL-C-19978.

Solderability: In accordance with MIL-C-19978.

Terminal strength: Method 211 of MIL-STD-202, test condition letter D.

Moisture resistance:

DWV:

Sleeving: In accordance with MIL-C-19978.

IR:

Sleeving: In accordance with MIL-C-19978.

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Stability at low and high temperature:

Low temperature:

Test temperature: -65°C +0°C, -5°C.

Capacitance change (maximum): -10 percent.

High temperature:

Test temperature: +125°C +5°C, -0°C.

Capacitance change (maximum: +10 percent.

Life: Capacitance change (maximum): ±5 percent of initial measured value.

Resistance to soldering heat: In accordance with MIL-C-19978.

Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, AT, AV, ME, MI
Navy - AS, MC, OS, SH, TD
Air Force - 17, 99

Preparing activity:
DLA - ES

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